

Secure Environment for Academic Assessments



Diogo Silva, Duarte Santos, Gabriel Janicas, Miguel Reis, Rodrigo Moço, Tiago Garcia

Table of contents



01

Our Project

02

Problems

03

Work and Results

04

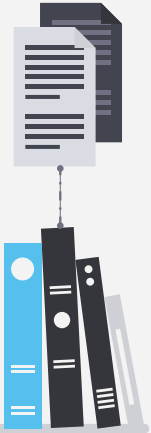
Architecture

05

Demonstration

06

Future Work





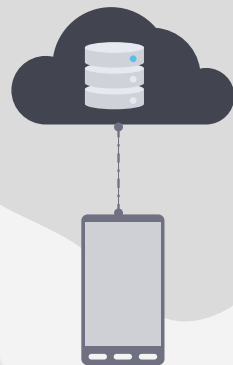
01

Our Project



What is our project ?

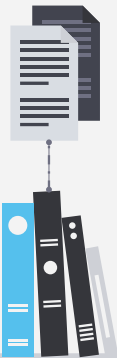
Create a Plug and Play
alternative to SEB, making it
available on all PC devices
without the need to install
any extra software.



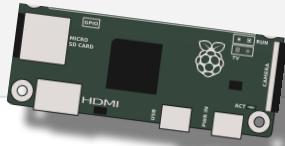
02

Problems

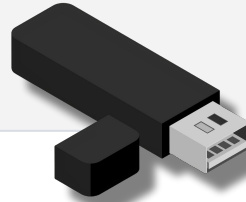




Problems

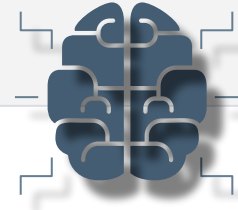


RAM consumption was going to be higher than the raspberry pi zero 2W could handle.

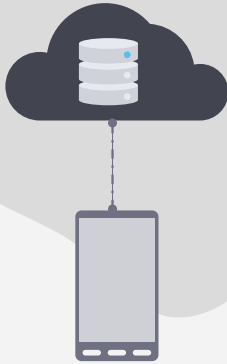


Solution: Use a Raspberry pi 4/5.

Problem: The cost of each device would already be very high



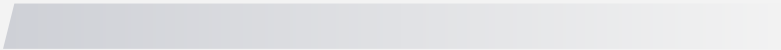
The gadget mode is not totally independent of the PC, it always needs an operating system to boot up.



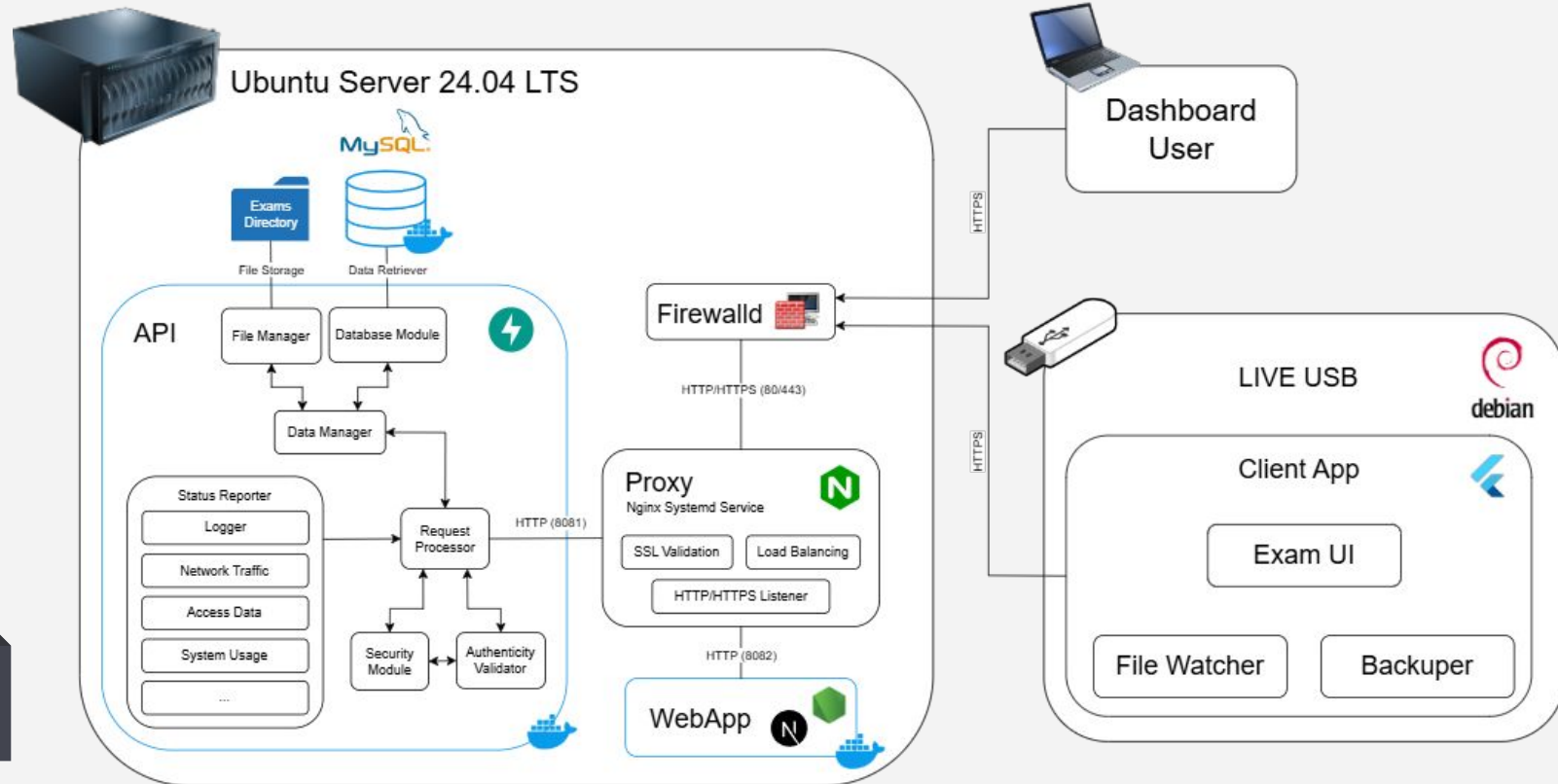


03

Architecture

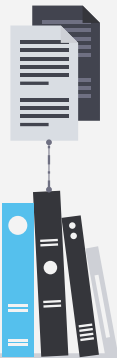


Architecture



04

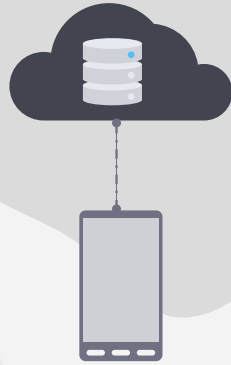
Work and Results

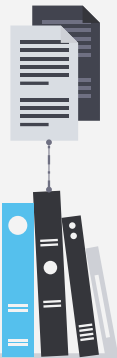


Client App



- Responsible for student authentication;
- Receives and **decrypts** the test to be carried out and delivers it back on submissions;
- Receives and manages possible **broadcast messages** sent by teachers (increase time, general notifications, etc);
- Receives and displays the remaining time of the current assessment.

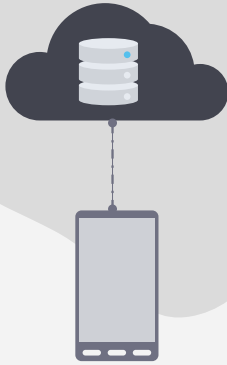




API



- Allows Professor-Student **Communication** in **real time**;
- Responsible for **securely** storing all data;
- Courses, Exams, Sessions and Users **Management**;
- Files and **Submissions Scanning**;



Web App



- Application created for **professors** to manage their assessments;
- Allows the professor to create **exams**, exam **sessions** and control the various sessions **simultaneously**;
- Allows you to send a **broadcast** message to **all** students or **individually**;



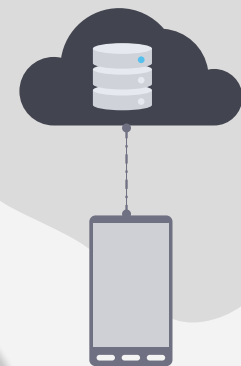




ISO



- Creation of a **Debian 12** image using **live-build**;
- Creation of two users, one for the **student** and one for the **administrators**;
- Set up a **Whitelist** on the **Linux firewall** (empty by default);
- Uploading the DevApp binaries along with the configuration file;



05

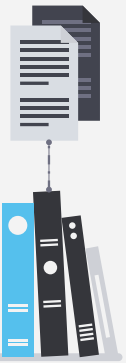
Demonstration



06

Future Work



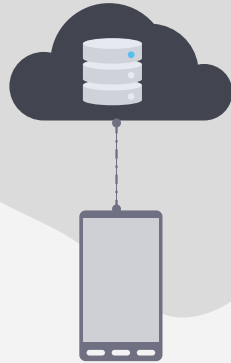


Future Work

- Improve the **security layers** in order to be trusted by any user, both teacher and student.

- Create a **role-based functionality** where we can have different users control the application in addition to the teacher and student.

- **Do a real simulation** of our system to show that our idea works correctly and to guarantee its credibility.



THANKS!

Do you have any questions?



<https://portableisolatedenvironment.github.io/>

